

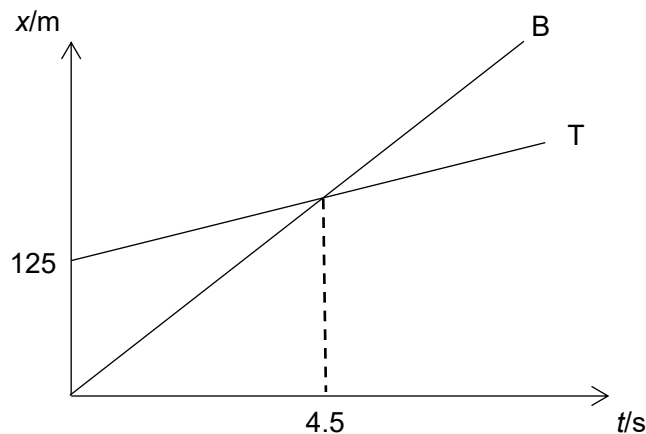
- 1 (a) The bomb is travelling at constant speed in the horizontal direction **B1**  
and is accelerating uniformly in the perpendicular (vertical) direction at a rate  
of  $9.81 \text{ m s}^{-2}$  (acceleration of free fall) **B1**

**MC:** Generally well answered. Most candidates were able to link their explanations to the definition of projectile motion and hence were able to attain full or minimally partial credit. Most who attained partial credit typically only stated that there is an acceleration in the vertical direction but did not explicitly mention that it is constant.

- (b)  $100 = 4.91 t^2$  **C1**  
 $t = 4.5 \text{ s}$  **A1**

**MC:** Generally well answered. Some candidate forgot to square in their working.

- (c) (i) 2 straight line graphs to show intersect at  $t = 4.5$  **A1**  
and when  $t = 0$ ,  $x_{\text{bomb}} = 0$  and  $x_{\text{truck}} = 125 \text{ m}$  **A1**  
No indication of values or graph labels: deduct 1 mark



**MC:** Decent attempt. Most candidates were able to obtain the graph above. Common errors included having both graphs start at the origin.

- (ii)  $v = \text{gradient of truck graph} = [(72 \times 4.5) - 125] / (4.5)$  **C1**  
 $= 44 \text{ m s}^{-1}$  **A1**

**MC:** Generally well answered. Ecf was should their answers in (b) or (c)(i) be wrong.

**2**

**2 (a) (i)** It states that the total momentum of a system of bodies is constant